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Retrieval Device System

Creating a system for retrieving devices from the bottom of the ocean is a very complex task. However, broken down, it becomes much easier to solve. Our problem to solve is being able to retrieve devices from the bottom of the ocean with these devices emitting some signal that is capable of being intercepted. Following this problem, there are 4 major systems that must be defined. The level 2, major systems are: Signal Detection, Signal Localization, Navigation, and Device Recovery.

To start, the signal detection system has two parts, signal receiving and processing. The signal receiving system has to receive the underwater signal via activating the receiver, and then confirming if the signal received is the desired signal. The signal processing in tandem with the signal receiver by removing any signal interference when the signal is confirmed, and then filtering the signal to be better understood by the next system in signal localization.

Signal localization also has two parts, finding direction and estimating position. When finding direction, the system measures which angle the signal is being intercepted at and then performs a separate task to further determine the direction toward the device. While this is happening, calculations are being conducted that determine where the device is compared to the vehicle, where then coordinates are given to the system that estimate where the device is, and finally the estimated position is confirmed and stored for future action.

Navigation, although sounds identical to signal localization, is more about moving the vehicle to the device rather than retrieving device position. The two parts to navigation are the navigation controller and depth sensor. Whenever the signal localization system is complete, the navigation system has time to activate. First, a navigation route is planned while the depth sensor is activated and starts keeping track of the depth of the vehicle. When the route is planned, the navigation system then guides the vehicle toward the device and then performs a separate action to adjust the vehicle in the proper position to retrieve the device.

Finally, the device recovery system’s parts consist of the device grabber and the lifting mechanism. After the other systems are complete and the vehicle is within reach of the device, the grabber goes in to grab the device, followed by further securing the device within itself to

ensure it does not fall out. Then, the lifting mechanism raises the device toward the surface, and it knows how far to ascend via calculations given by the depth sensor, followed by further initiating ascent of the device to be fully recovered.

